SCTR's Pune Institute of Computer Technology Dhankawadi, Pune

AN INTERNSHIP REPORT ON

Title : "End-to-End Development of Role-Based Mobile and Web Applications Across Multiple Industry Domains"

SUBMITTED BY

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Class: TEXI

Roll no: 33347

Under the guidance of

Mrs. Swapnaja R. Hiray

DEPARTMENT OF INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025-26

DEPARTMENT OF INFORMATION TECHNOLOGY

SCTR’s Pune Institute of Computer Technology
Dhankawadi, Pune
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CERTIFICATE

This is to certify that the SPPU Curriculum-based internship report entitled

“End-to-End Development of Role-Based Mobile and Web Applications Across Multiple Industry Domains”

Submitted by

Vivek Jitendra Patil
Roll no : 33347

has satisfactorily completed the curriculum-based internship under the guidance of Mrs. Swapnaja R. Hiray towards the partial fulfillment of third year Information Technology Semester VI, Academic Year 2025-26 of Savitribai Phule Pune University.

Mrs. Swapnaja R. Hiray	**(Dr.) Emmanuel M.**
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	PICT, Pune

Place: Pune

Date:

ACKNOWLEDGEMENT

I would like to express my gratitude and appreciation to all those who gave me the possibility to complete this report. Special thanks is due to my Guide Mrs. Swapnaja R. Hiray and Reviewer Mr. Vinit R. Tribhuvan whose help, stimulating suggestions and encouragement helped me in all time of fabrication process and in writing this report. I also sincerely thanks for the time spent proof- reading and correcting my many mistakes

I would also like to thank my Internship Mentor Mrs. Swapnaja R. Hiray to continuously help me through the course of making this report and also providing resources to complete the tasks.

Many thanks to the PICT faculty and Dr. Emmanuel M, head of information technology department for providing me the opportunity to complete my Internship and present this report by providing all the resources required.

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Roll No: 33347

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1. Introduction

The field of software engineering has witnessed a remarkable transformation over the past decade, with organisations across every industry adopting digital solutions to streamline their operations, improve efficiency, and enhance customer experiences. The demand for full-stack developers proficient in both mobile and web platforms has grown exponentially, making hands-on industry exposure an essential component of technical education.

This internship was undertaken at CustomSoft, Pune, as part of the SPPU Curriculum-Based Internship Programme for the Third Year Information Technology, Semester VI, Academic Year 2025-26, under the esteemed guidance of Mrs. Swapnaja R. Hiray. The internship spanned from January 2026 to April 2026, covering a period of approximately three and a half months of intensive, project-based learning.

During this internship, four distinct software projects were designed, developed, and deployed, each targeting a different industry domain. These projects were built using a diverse set of modern technologies including ASP.NET Core MVC, Flutter, Node.js, and React. Each application followed a role-based architecture, ensuring that different types of users had appropriate access and functionality tailored to their responsibilities.

The four projects developed during this internship are as follows:

- Student Teacher Portal — A web-based student management and academic portal developed using ASP.NET Core MVC, Entity Framework Core, and SQL Server, supporting Teacher and Student roles.

- Business ERP System — A comprehensive multi-role Enterprise Resource Planning platform built using Node.js and Flutter, catering to Admin, Sales Manager, Planning Manager, Purchase Manager, Client, and Vendor roles across the full business lifecycle.

- Hotel Management System — A Flutter-based hotel operations and sales management application designed for Admin, Booking Manager, and Sales Executive roles, handling room management, bookings, payments, and analytics.

- FixItPro Urban Services Platform — A service booking platform built with Node.js and React, connecting Users (customers), Service Providers (technicians/professionals), and Admin for managing on-demand urban services.

Each of these projects demanded a thorough understanding of database schema design, REST API development, mobile UI/UX design, role-based access control, real-time notifications, location-based services, and deployment workflows. The internship provided an invaluable opportunity to apply theoretical concepts from the academic curriculum to practical, real-world problems and to work collaboratively as part of a professional development team.

This report documents the objectives, methodology, tools, and outcomes associated with the internship work undertaken. It reflects the technical skills acquired, the challenges faced, and the solutions developed over the course of this enriching professional experience.

2. Problem Statement

Modern businesses and service organisations face numerous operational challenges stemming from fragmented, manual, or outdated systems. The absence of integrated, role-based digital platforms leads to inefficiencies in communication, data management, and process coordination. Across four distinct industry domains — education, manufacturing and sales, hospitality, and urban services — the following critical problems were identified:

2.1 Education Sector — Student Teacher Portal

Educational institutions continue to manage student records using manual registers, spreadsheets, or basic systems that lack automation. Key challenges include:

- No centralised platform to manage student profiles, academic records, and parent details in one place.
- Manual and time-consuming student data entry, with no provision for bulk import via CSV.
- Absence of role-based login, leading to data privacy concerns where students can access administrative data.
- No automated notification mechanism when student records are added, updated, or deleted.
- Inability to generate and export student reports in structured formats such as CSV or PDF.

2.2 Manufacturing & Sales — Business ERP System

Manufacturing and sales companies often operate with disconnected teams spread across sales, production planning, procurement, and client management. The identified challenges are:

- No single unified platform connecting Sales Managers, Planning Managers, Purchase Managers, Clients, and Admin.

- Business workflows such as Enquiry → Quotation → LOI → Invoice → Payment tracking were managed manually, leading to delays and errors.
- Lack of real-time chat and notification features caused communication gaps between internal teams and external clients.
- No mechanism for tracking production components, inventory status, or work orders digitally.
- Multi-role access was absent, making it difficult to assign role-specific responsibilities and data visibility.

2.3 Hospitality Sector — Hotel Management System

The hotel and hospitality industry relies heavily on efficient booking management, agent coordination, and sales tracking. Common problems include:

- No centralised dashboard for managing hotel master data, room details, meal plans, and agent information.
- Booking creation, modification, tracking, and payment management were handled in silos without a unified system.
- Sales Executives lacked a structured platform to record meetings, follow-ups, visit reports, and action points.
- Business analytics such as occupancy rates, revenue summaries, and booking trends were not available in real time.
- Absence of role-based access meant all staff members had unrestricted access to sensitive operational data.

2.4 Urban Services — FixItPro Home Services Platform

The on-demand home services sector faces significant challenges in connecting customers with skilled service providers in a reliable, transparent, and location-aware manner:

- Customers had no reliable digital platform to browse services, book appointments, and track service status in real time.
- Service providers lacked a structured registration, verification, and job-assignment mechanism.

- Admin had no centralised control panel to manage service categories, pricing, slots, provider approvals, and booking assignments.
- OTP-based secure login and online payment integration were absent in existing local service booking solutions.
- Location-based logic for finding nearby service providers was not implemented in legacy systems.

In summary, each of these domains suffered from the lack of a well-architected, role-based, and integrated digital solution. This internship aimed to address these gaps through the end-to-end design, development, and deployment of custom software applications tailored to each domain.

3. Objectives and Scope

3.1 Objectives

The primary objectives of this internship were defined at the outset to ensure focused, meaningful, and industry-relevant learning outcomes. The objectives are listed as follows:

1. To gain practical experience in end-to-end software development by designing, building, and deploying real-world applications across multiple domains.
2. To implement role-based access control (RBAC) architectures in both mobile and web applications, ensuring that each user role has appropriate permissions and data visibility.
3. To develop proficiency in a diverse technology stack encompassing ASP.NET Core MVC, Flutter, Node.js, and React, along with databases such as SQL Server, Firebase, and Supabase.
4. To design normalised and scalable database schemas that support complex multi-role business workflows and relational data structures.
5. To integrate backend REST APIs with Flutter mobile frontends and React web frontends, enabling seamless data exchange and state management.
6. To implement advanced application features such as real-time notifications, OTP-based authentication, location-based service discovery, online payment integration, file upload/download, and CSV/PDF export.
7. To understand and practise collaborative software development workflows, including version control using Git, code review, module integration, and iterative feature development.
8. To deploy backend services and web frontends on cloud hosting platforms, gaining exposure to production-level deployment processes.
9. To develop professional soft skills including project communication, documentation, task planning, and problem-solving under real-world time constraints.

3.2 Scope

The scope of this internship covered the complete software development lifecycle — from requirements gathering and database schema design to implementation, testing, and deployment — for four distinct projects. The scope is detailed below:

Student Teacher Portal: The scope included development of a full-featured student management dashboard with CRUD operations, profile photo upload, CSV import/export, PDF report generation, role-based login (Admin and Student), and automated notifications. The system was built as a web application using ASP.NET Core MVC.

Business ERP System: The scope covered the complete business operations workflow for a manufacturing and sales company. This included multi-role user management (6 roles), end-to-end business flow from enquiry to payment, production planning with components and work orders, real-time chat and notification features, vendor management, and admin-level analytics. The system included both a Flutter mobile application and a Node.js backend.

Hotel Management System: The scope encompassed hotel master data setup, room and meal management, agent coordination, booking creation and tracking, payment management, sales executive visit and follow-up tracking, and real-time business analytics. The application was built as a Flutter mobile application.

FixItPro Urban Services Platform: The scope included OTP-based user and provider authentication, service category and product management by Admin, user-side service browsing and slot-based booking, online payment processing, provider registration and verification, job assignment, real-time notifications, and location-based radius configuration for provider discovery. The platform was built using Node.js for the backend and React for the web frontend.

The scope did not include the development of advanced AI/ML features, hardware integrations, or enterprise-level scalability testing. It was limited to the features and modules agreed upon with the internship mentor and relevant for the curriculum-based internship objectives.

4. Methodological Details

4.1 Designing and Developing Role-Based Mobile and Web Applications

The development methodology followed during this internship was iterative and project-driven, closely resembling the Agile software development approach. Each project was broken into phases: requirement analysis, schema design, UI/UX design, backend development, frontend development, integration, testing, and deployment. The following subsections describe the design and development approach for each of the four projects.

4.1.1 Student Teacher Portal (.NET)

The Student Teacher Portal was the first project undertaken during the internship. It was developed as an ASP.NET Core MVC web application with Entity Framework Core as the ORM and Microsoft SQL Server as the database.

The database schema was designed to capture all essential student-related information including personal details (name, date of birth, gender, address), academic details (roll number, course, admission year), parent/guardian details (father name, mother name, contact numbers), and a profile photograph stored as a file path. A separate Notifications table was designed to log all system events related to student records.

The admin-side functionality was built as a comprehensive student management dashboard featuring:

- Full CRUD operations — adding new students, viewing student lists with pagination, editing existing records, and deleting students.
- Advanced filtering and search by course, gender, and keyword across student records.
- Profile photo upload and storage with server-side file handling.
- Bulk student data import from CSV files with field mapping and validation.
- Export of student records as CSV and PDF reports for administrative use.
- Automatic notification creation on every student add, edit, or delete action, with unread count display and mark-as-read functionality.

The student-side functionality provided each student with a personalised profile page displaying their personal details, course information, parent details, and profile photograph after login. A basic authentication system with role-based redirection was implemented — Admin users were redirected to the management dashboard while Student users were redirected to their individual profile page.

4.1.2 Business ERP System (React, Node.js & Flutter)

The Business ERP System was the most complex project of the internship, involving six distinct user roles and a complete business workflow from lead enquiry to final payment. The backend was developed using Node.js with Express.js and initially used Firebase as the database, which was later migrated to Supabase (PostgreSQL) for improved scalability and structured querying.

The database schema was designed through collaborative sessions and covered entities such as Companies, Users (multi-role), Clients, Products, Enquiries, Quotations, LOIs (Letter of Intent), Invoices, Payments, Components, Work Orders, Chat Messages, and Notifications. Relationships between these entities were carefully normalised to support the end-to-end business workflow.

The end-to-end business flow implemented in the system is as follows:

- Admin sets up the company profile, creates team member accounts (Sales, Planning, Purchase Managers), and manages the client and product master data.
- Client raises an enquiry through the client-side application, specifying product requirements and quantities.
- Sales Manager reviews the enquiry, creates a detailed quotation, and sends it to the Client.
- Client reviews the quotation and submits an LOI (Letter of Intent) with approval or rejection, along with comments.
- Upon LOI approval, Sales Manager proceeds to generate an invoice and initiates payment follow-up.
- Planning Manager handles the production side — defining components, updating inventory status, and creating/tracking work orders for production fulfilment.
- All parties receive real-time notifications for key events; live chat enables direct communication between roles.

The Flutter mobile application was built with a modular drawer-based navigation system. Each role had a separate, role-specific drawer with access only to the modules relevant to that role. The UI screens were developed iteratively, beginning with the Sales Manager side, followed by the Client side, Admin side, and subsequently the Planning Manager, Purchase Manager, and Vendor modules. Admin-side analytics dashboards provided high-level business metrics. A profile image feature was added to all role profiles.

A significant technical challenge addressed during this project was the integration of two separate backend login systems (originating from different team members) into a unified authentication flow. Additionally, the migration from Firebase to Supabase required re-designing the data access layer and updating all API endpoints accordingly.

4.1.3 Hotel Management System (React, Node.js & Flutter)

The Hotel Management System was developed as a Flutter mobile application with a backend supporting three primary roles: Admin, Booking Manager (also referred to as Sales Manager in parts of the backend), and Sales Executive.

The database schema was designed to handle hotel master data (hotel details, rooms, room types, meal plans), agent information, bookings (with full stay details, check-in/check-out dates, room assignments), restaurant bookings, payments (advance, balance, receipt tracking), and sales-side records (meetings, visits, follow-ups, notes, action points, reports).

The Admin module provided master setup capabilities including management of hotels, agents, rooms, meals, users, and access to high-level business analytics. The Booking Manager module was the operational core of the application — it allowed for booking creation, editing, CSV import of bookings, status tracking, and payment management. The Sales Executive module focused on the field operations side: logging meetings, recording visits, setting follow-up reminders, writing notes and action points, and generating sales reports.

The application was tested end-to-end during the internship period, with screen corrections applied to align the UI with the agreed design specifications.

4.1.4 FixItPro Urban Services Platform (Node.js & React)

FixItPro is a service booking platform inspired by urban home services aggregators. The platform connects customers who need repair and maintenance services with nearby service providers (technicians and professionals), with Admin overseeing the entire operation. The backend was developed using Node.js and the customer/admin web interface was built using React.

Key features developed for each role are as follows:

Admin: Service category and product management with pricing, slots, and service details configuration. Provider submission review with approval/rejection workflow. Full booking visibility with provider assignment capability. Location-based logic for finding providers within a defined radius. Notification monitoring.

User (Customer): OTP-based secure login via SMS. Service browsing by category with slot selection. Booking creation, status tracking, and cancellation. Online payment processing. Service rating upon completion. Profile management.

Provider (Service Professional): Registration with document submission for admin verification. Personalised dashboard with real-time analytics (total jobs, earnings, ratings). Assigned job viewing and status updates. Real-time push notifications for new job assignments. Radius configuration to define the service area. Profile and document management.

The provider-side application had the most feature-rich development phase during this project. Key technical implementations included Google Maps integration for location-based services, real-time analytics on the provider dashboard, notification integration using Firebase Cloud Messaging, detailed service overview screens, complete profile management, and a redesigned booking flow that improved the provider experience significantly.

4.2 Deploying Multi-Module Applications Across Industry Domains

Deployment was an integral part of the internship experience, providing exposure to the complete software delivery lifecycle beyond just development. The following deployment activities were carried out:

- The Node.js backend for the Business ERP System was hosted on a cloud backend hosting platform during the internship period. This involved setting up environment variables, configuring the production database connection to Supabase, and validating API endpoints post-deployment.
- The ERP web dashboard (Admin and Sales Manager web views) was hosted as a static web application, with screen corrections applied post-hosting to align with the production design specifications.
- A demo website for a new project concept was created and published using Lovable, a rapid web development and hosting tool, demonstrating the ability to use modern low-code/no-code deployment tools alongside traditional development approaches.
- The FixItPro React web application was deployed and made accessible, with post-deployment testing carried out to verify booking flows, notification delivery, and provider-side operations.
- Towards the end of the internship, a WordPress setup was initiated, providing exposure to CMS-based deployment for content-driven web projects.

Throughout the deployment process, the intern gained practical knowledge of hosting configurations, environment management, API endpoint validation on production, cross-origin resource sharing (CORS) configuration, and debugging deployment-specific issues that do not arise in local development environments.

5. Modern Engineering Tools Used

The internship involved the use of a wide range of modern engineering tools, frameworks, and platforms. These are categorised and described below:

Category	Tool / Technology	Purpose
Frontend Framework	ASP.NET Core MVC	Web application development for Student Teacher Portal
Frontend Framework	React.js	Web frontend for FixItPro Urban Services Platform
Mobile Framework	Flutter (Dart)	Cross-platform mobile development for ERP and Hotel Management
Backend Runtime	Node.js with Express.js	REST API development for ERP and Urban Services platforms
ORM / Data Access	Entity Framework Core	Database access layer for ASP.NET Core application
Database	Microsoft SQL Server	Relational database for Student Teacher Portal
Database	Firebase (Firestore)	Initial NoSQL database for ERP System (later migrated)
Database	Supabase (PostgreSQL)	Production database for ERP System post-migration
Version Control	Git & GitHub	Source code management and team collaboration
IDE / Editor	Visual Studio / VS Code	Primary development environments
IDE	Android Studio	Flutter mobile development and emulation
API Testing	Postman	REST API testing and documentation
Deployment	Cloud Backend Hosting	Node.js backend deployment for ERP system
Deployment	Lovable	Rapid web app deployment for demo project
Deployment	WordPress	CMS-based web deployment (setup initiated)
Push Notifications	Firebase Cloud Messaging	Real-time push notifications for mobile apps

Category	Tool / Technology	Purpose
Maps & Location	Google Maps API	Location-based services in FixItPro platform
Authentication	OTP via SMS Service	Secure login for FixItPro users and providers
Report Generation	iTextSharp / PDF Library	PDF report export in Student Teacher Portal
Database Migration	Supabase Migration Tools	Firebase to Supabase data and schema migration

The combination of these tools provided a comprehensive, industry-relevant development environment that closely mirrors the toolchains used in professional software development organisations. Working across multiple platforms — web, mobile, and backend — within a single internship period significantly broadened the intern's technical perspective and adaptability.

6. Outcome / Result of Internship Work

The internship at CustomSoft resulted in the successful design, development, and partial deployment of four fully functional, role-based software applications across different industry domains. The outcomes are summarised below:

6.1 Technical Outcomes

- Successfully developed and delivered a Student Teacher Portal with full CRUD, import/export, notifications, and role-based login using ASP.NET Core MVC.
- Built a comprehensive multi-role Business ERP System with end-to-end business flow, real-time chat, notifications, and admin analytics using Node.js and Flutter.
- Developed a Hotel Management System with booking, payment, analytics, and sales tracking modules using Flutter.
- Created the FixItPro Urban Services Platform with OTP login, real-time provider dashboard, maps integration, notification system, booking flow, and online payment using Node.js and React.
- Gained hands-on experience with database design across SQL (SQL Server, Supabase/PostgreSQL) and NoSQL (Firebase) paradigms.
- Successfully performed a live database migration from Firebase to Supabase for the ERP system in a production context.
- Deployed backend APIs and web frontends to production cloud hosting environments.

6.2 Project-Wise Summary

Project	Tech Stack	Roles	Key Features Delivered
Student Teacher Portal	.NET Core MVC, SQL Server	Admin, Student	CRUD, CSV import/export, PDF report, photo upload, notifications, role login
Business ERP System	Node.js, Flutter, Supabase	Admin, Sales Mgr, Planning Mgr, Purchase Mgr, Client, Vendor	Enquiry-to-payment flow, real-time chat, notifications, work orders, analytics dashboard
Hotel Management System	Flutter	Admin, Booking Manager, Sales Executive	Booking management, payment tracking, agent management, sales reports, analytics
FixItPro Urban Services	Node.js, React	Admin, User, Provider	OTP login, maps, slot booking, online payment, provider verification, real-time notifications

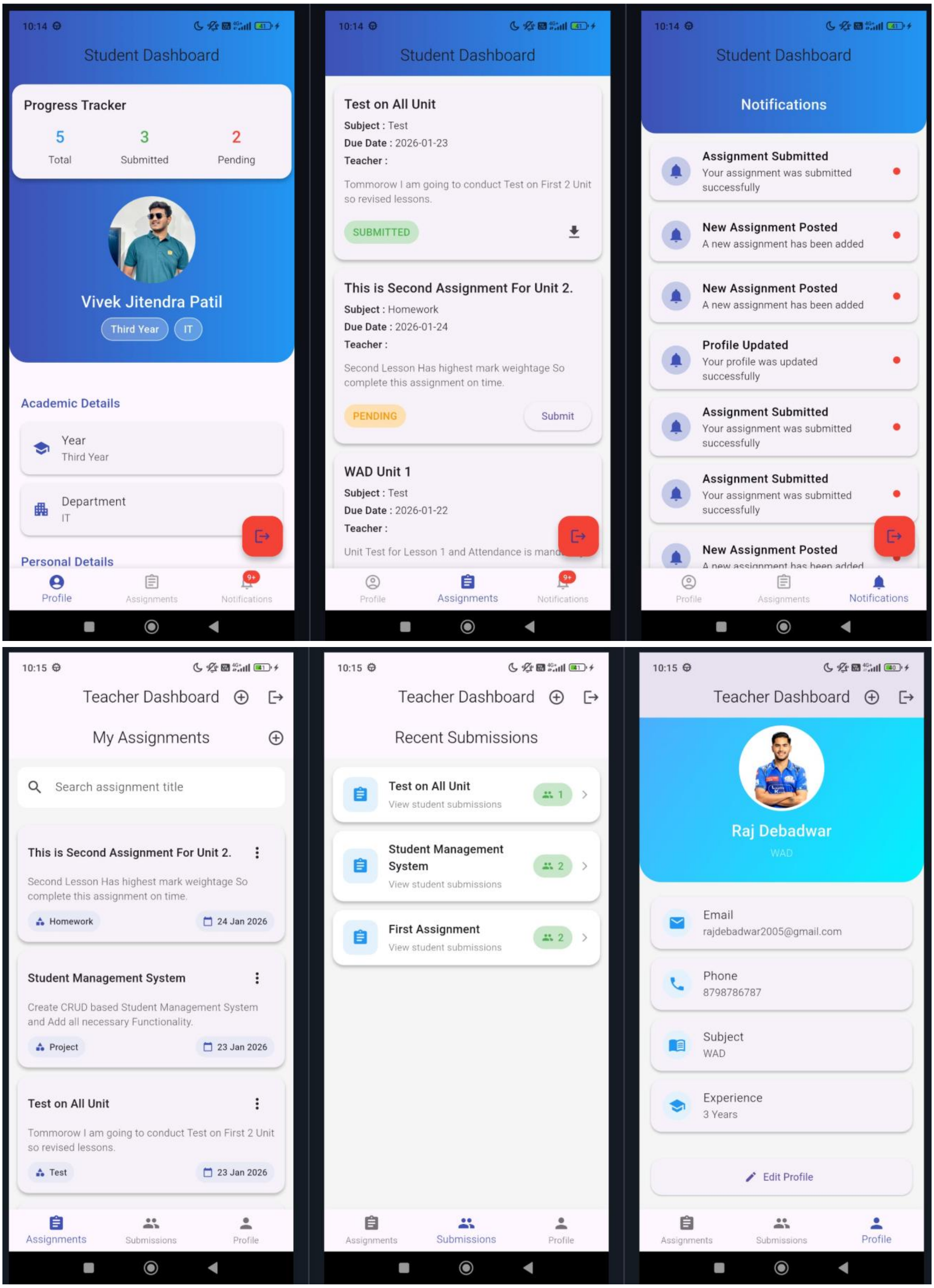
6.3 Learning Outcomes

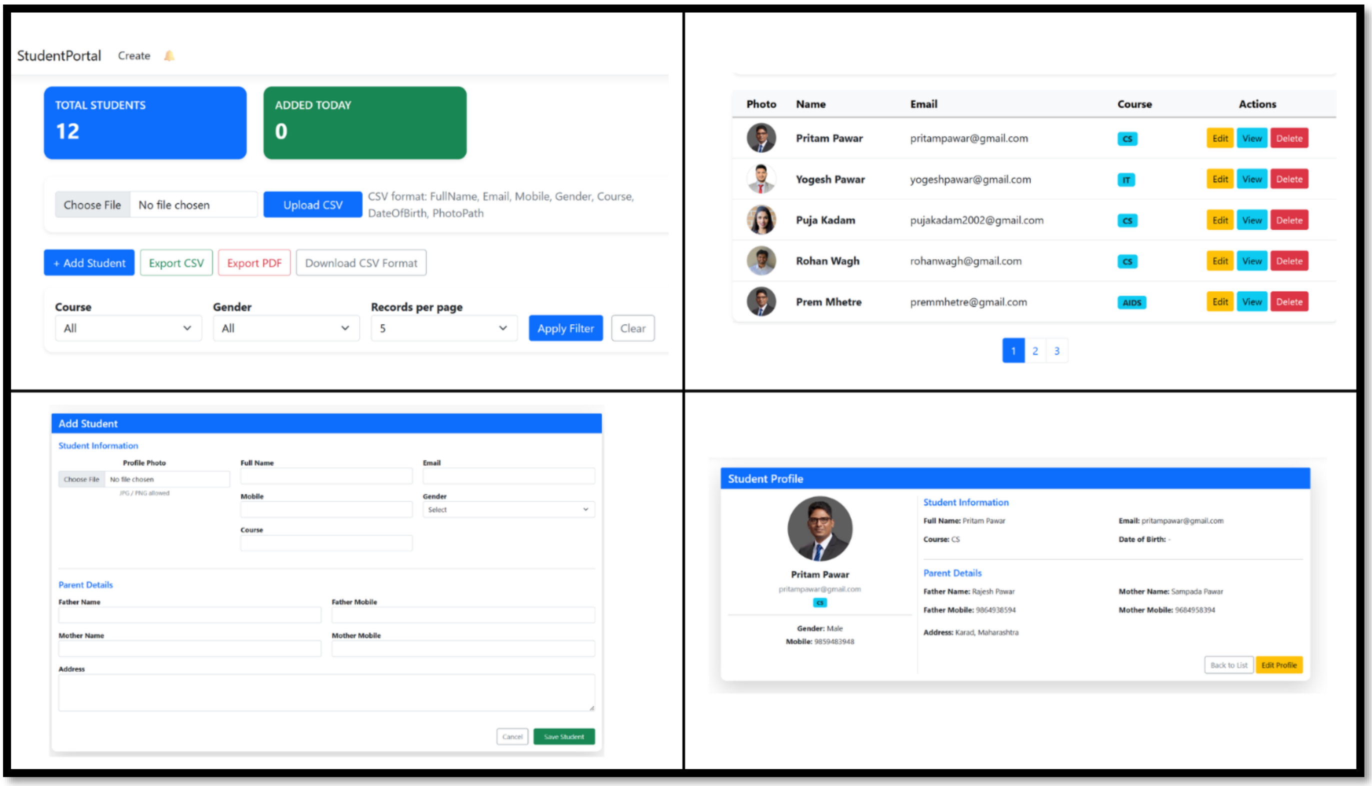
- Developed strong proficiency in cross-platform mobile development with Flutter and Dart, including complex multi-role navigation and state management patterns.
- Gained confidence in building RESTful APIs using Node.js and Express.js with authentication, role-based middleware, and third-party service integrations.
- Learned to design and implement normalised relational database schemas for complex, multi-entity business applications.
- Acquired practical experience in software integration — merging modules developed by different team members, resolving conflicts, and ensuring seamless end-to-end functionality.
- Developed an understanding of the full software deployment lifecycle, including environment configuration, API validation, and debugging production-specific issues.
- Enhanced professional skills including collaborative development using Git, task planning, technical documentation, and effective communication within a team setting.

6.4 Screenshots of Project Outputs

The following pages contain screenshots from the developed applications, showcasing the user interfaces, dashboards, and key features implemented across all four projects.

Project 1:




StudentPortal
Create
TOTAL STUDENTS
12
ADDED TODAY
0
Choose File
No file chosen
Upload CSV
CSV format: FullName, Email, Mobile, Gender, Course, DateOfBirth, PhotoPath
+ Add Student
Export CSV
Export PDF
Download CSV Format
Course
Gender
Records per page
Apply Filter
Clear
Photo
Name
Email
Course
Actions
Pritam Pawar
pritampawar@gmail.com
Yogesh Pawar
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Puja Kadam
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Rohan Wagh
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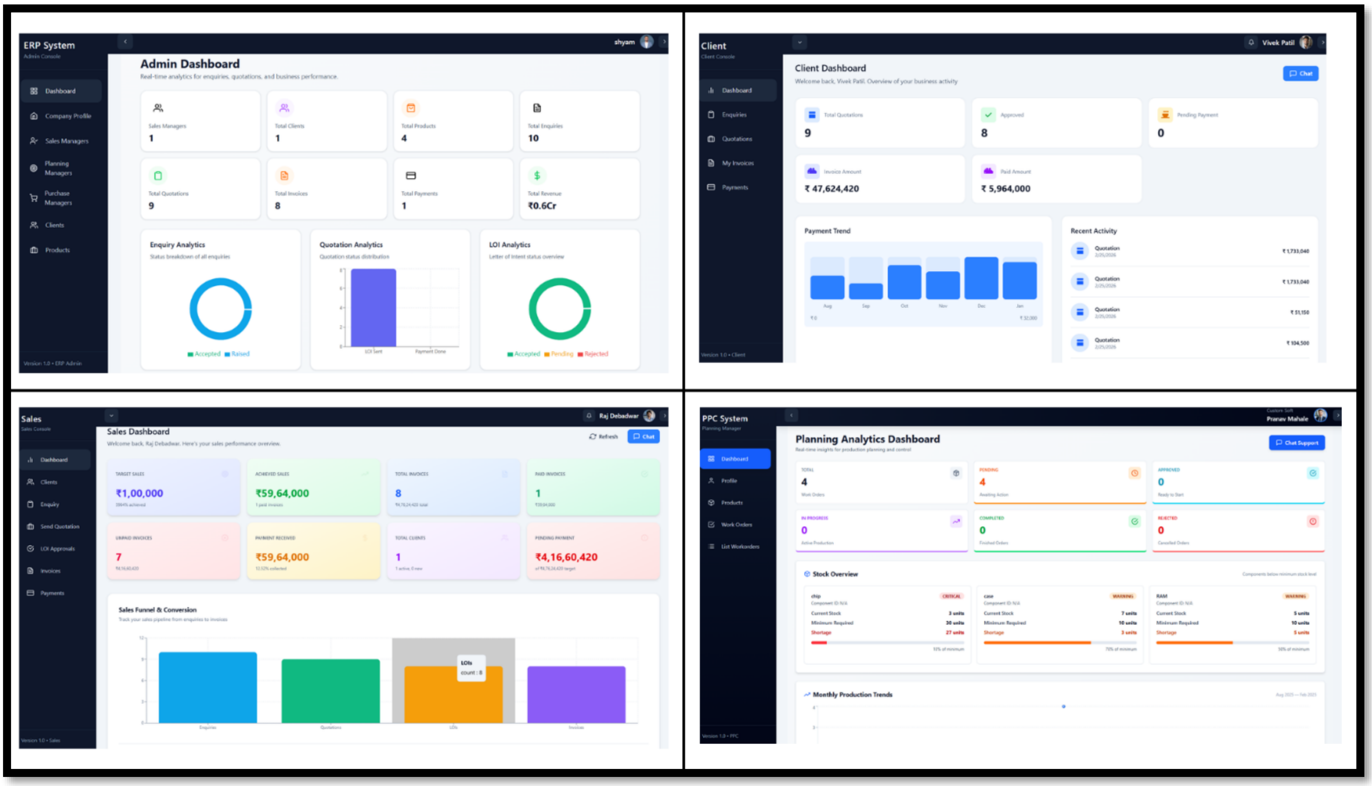
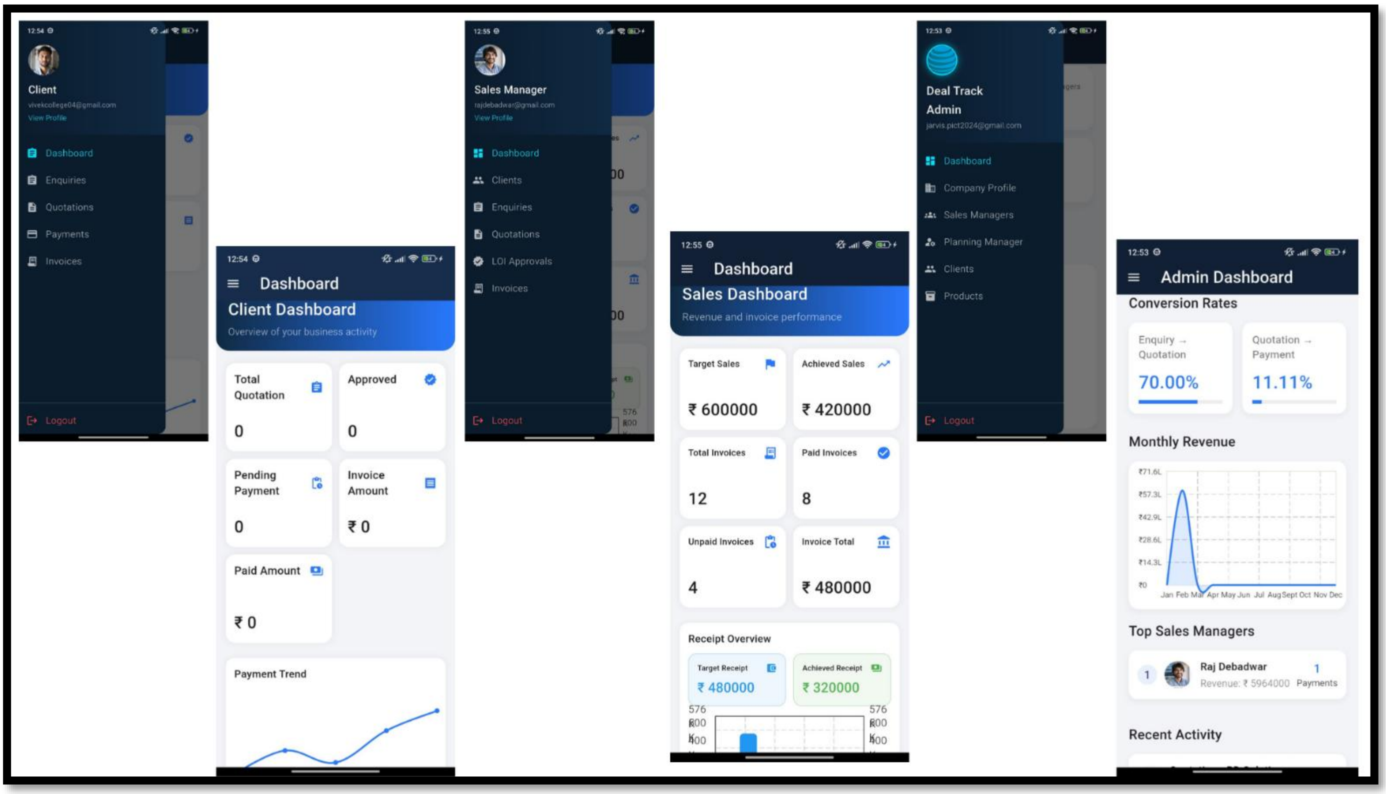
Edit
View
Delete
Add Student
Student Information
Parent Details
Student Profile
Pritam Pawar

Project 2:


ERP System
Admin Dashboard
Client Dashboard
Sales Dashboard
Planning Analytics Dashboard
PPC System


Client
Dashboard
Enquiries
Quotations
Payments
Invoices
Logout
Dashboard
Client Dashboard
Overview of your business activity
Total Quotation
0
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Pending Payment
0
Invoice Amount
₹ 0
Paid Amount
₹ 0
Payment Trend
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Dashboard
Clients
Enquiries
Quotations
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Receipt Overview
Target Receipt
₹ 480000
Achieved Receipt
₹ 320000
Deal Track
Admin
Dashboard
Company Profile
Sales Managers
Planning Manager
Clients
Products
Logout
Admin Dashboard
Conversion Rates
Enquiry → Quotation
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Quotation → Payment
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Top Sales Managers
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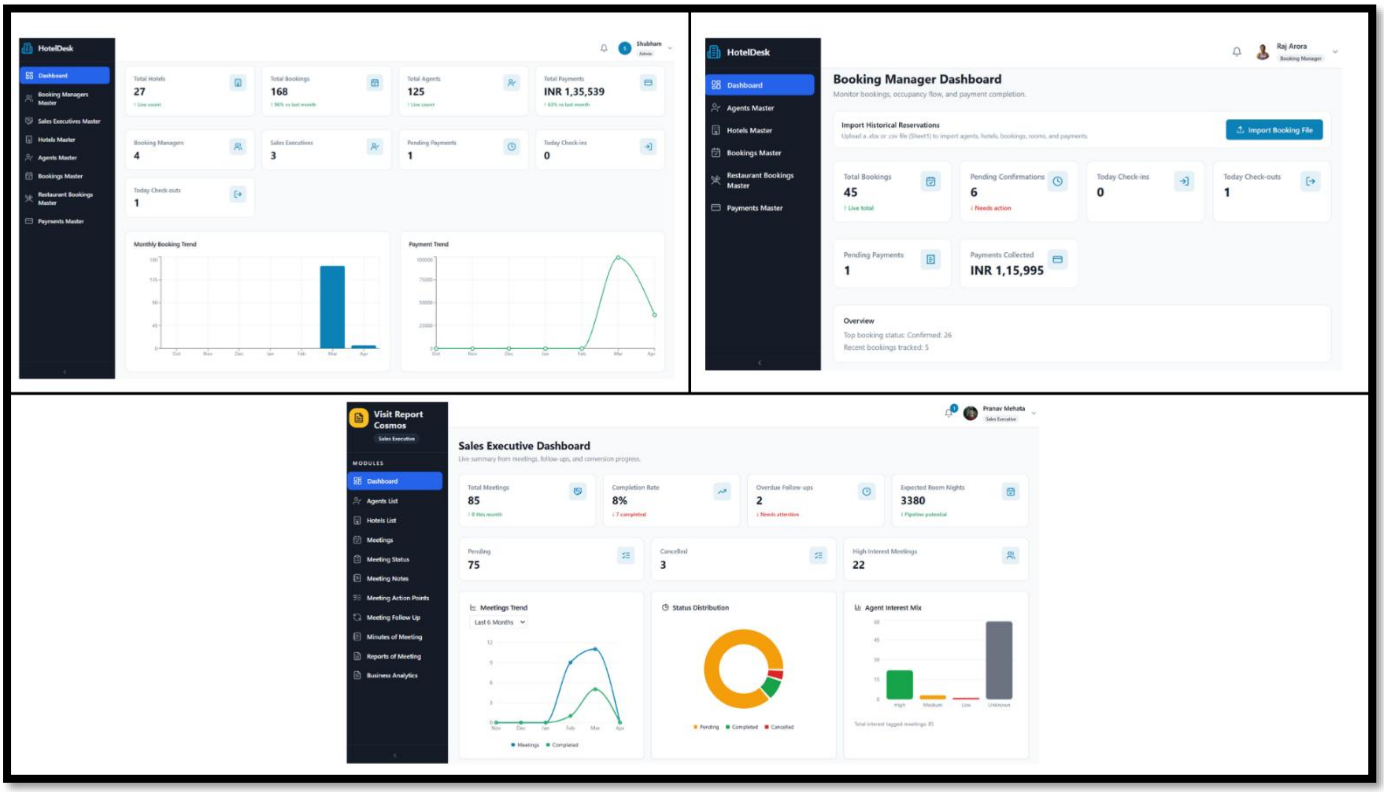
Payments
Recent Activity

Project 3:


HotelDesk
Booking Manager Dashboard
Import Historical Reservations
Import Booking File
Total Bookings
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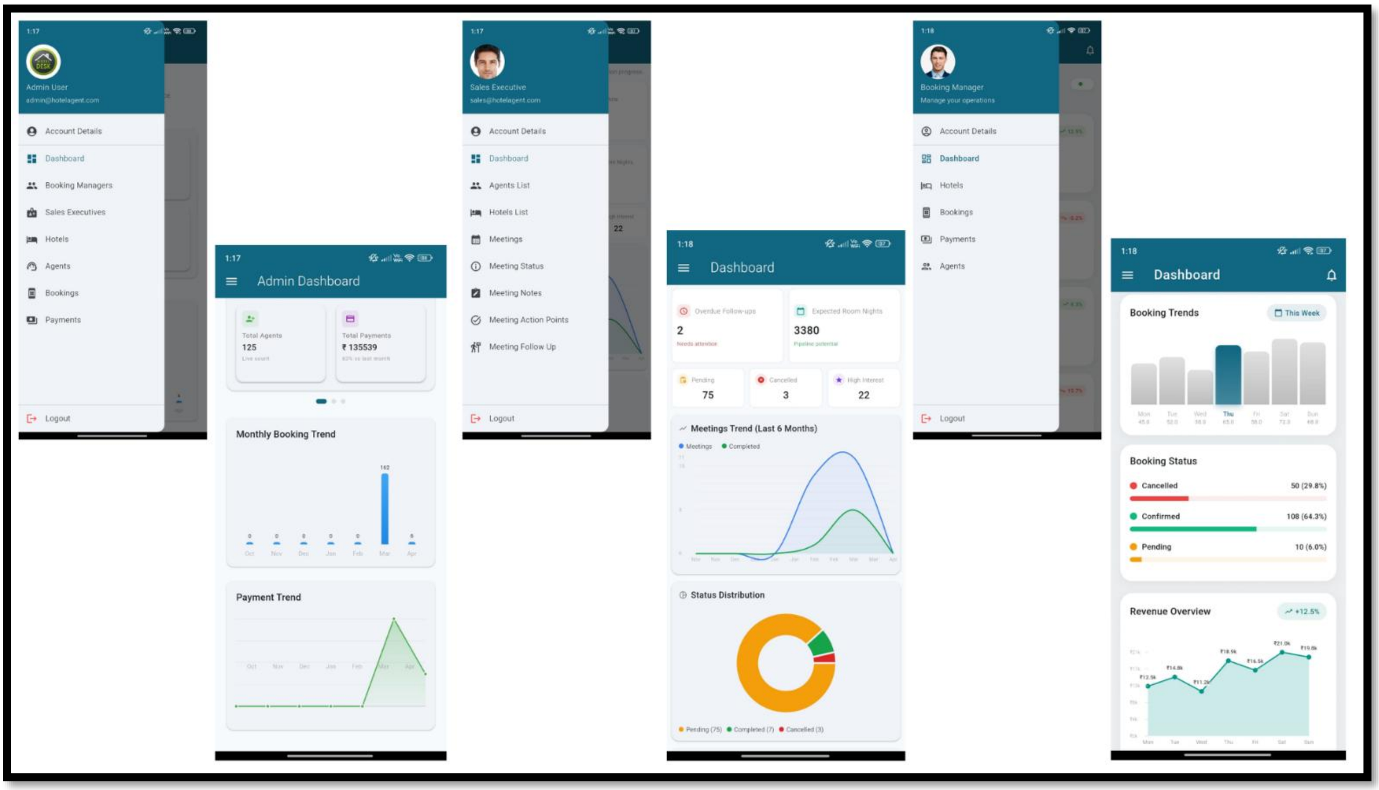
Pending Confirmations
6
Pending Payments
1
Payments Collected
INR 1,15,995
Overview
Visit Report Cosmos
Sales Executive Dashboard
Meetings Trend
Status Distribution
Agent Interest Mix



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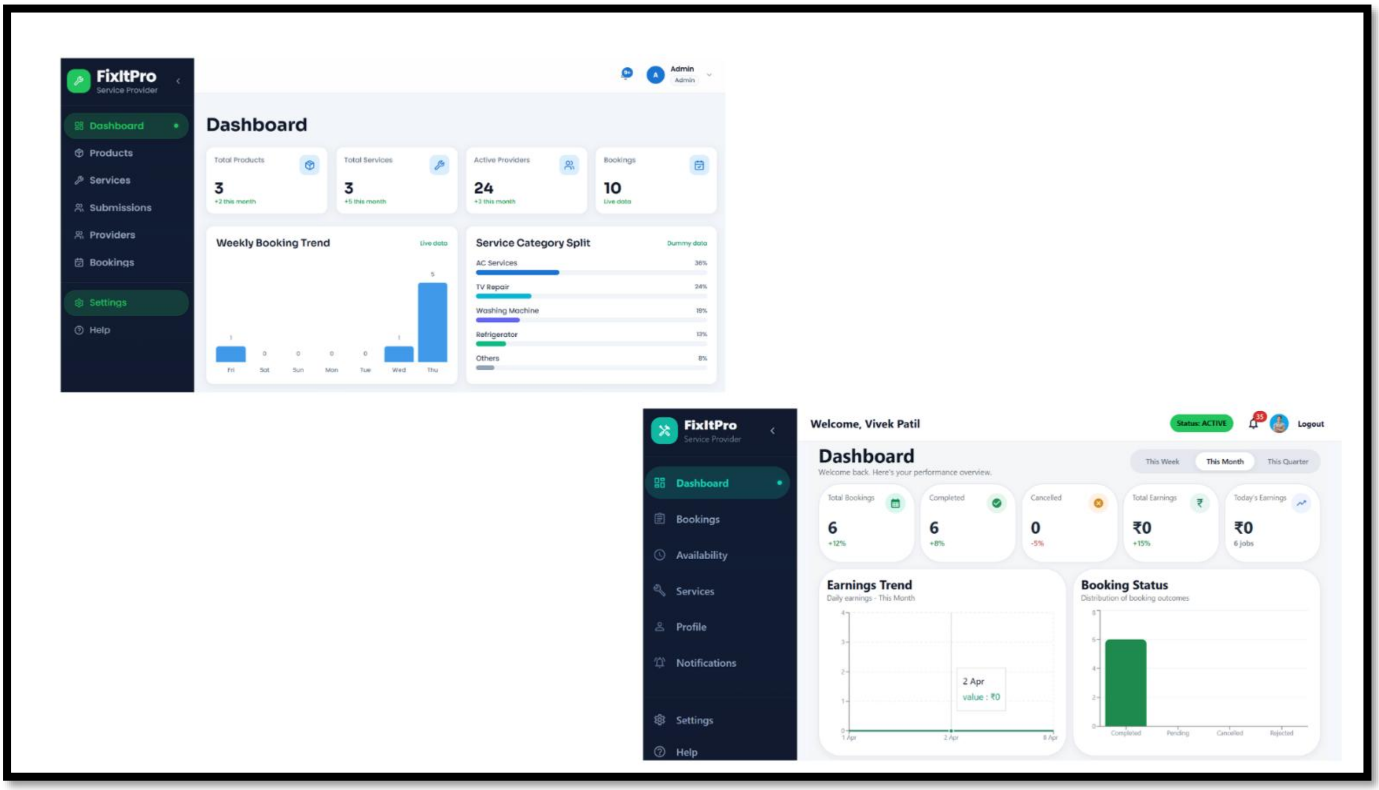

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